Network Infrastructure & pfSense Configuration

Overview

This document details the complete network infrastructure setup using **pfSense** as the core firewall and routing platform, implementing enterprise-grade **VLAN segmentation** with a **TP-Link TL-SG108E managed switch**. The architecture provides isolated network segments for different security functions while maintaining centralized control and monitoring.

Hardware Architecture

Core Components

- **pfSense Firewall**: Dedicated desktop machine with 2 NICs
- **TP-Link TL-SG108E (Switch 1)**: 8-Port Gigabit Smart Managed Switch; Port 1 trunk to pfSense, Port 2 trunk to Switch 2
- **Switch 2 (secondary switch)**: uplinked from Switch 1 Port 2; passes the tagged VLANs through to Proxmox
- **Primary Laptop**: Administrative workstation (`192.168.10.3`)
- **Proxmox VE Host**: Bare-metal hypervisor connected to Switch 2 (`192.168.10.6`), receiving all tagged VLANs
- **Vault Desktop**: Dedicated machine for HashiCorp Vault on Port 6, VLAN 40 (`192.168.40.2`, awaiting configuration)
- **Windows Server 2025 Host**: Dedicated hardware on Port 7, VLAN 50, domain controller at `192.168.50.2` (connected via USB-to-Ethernet adapter)

Physical Topology

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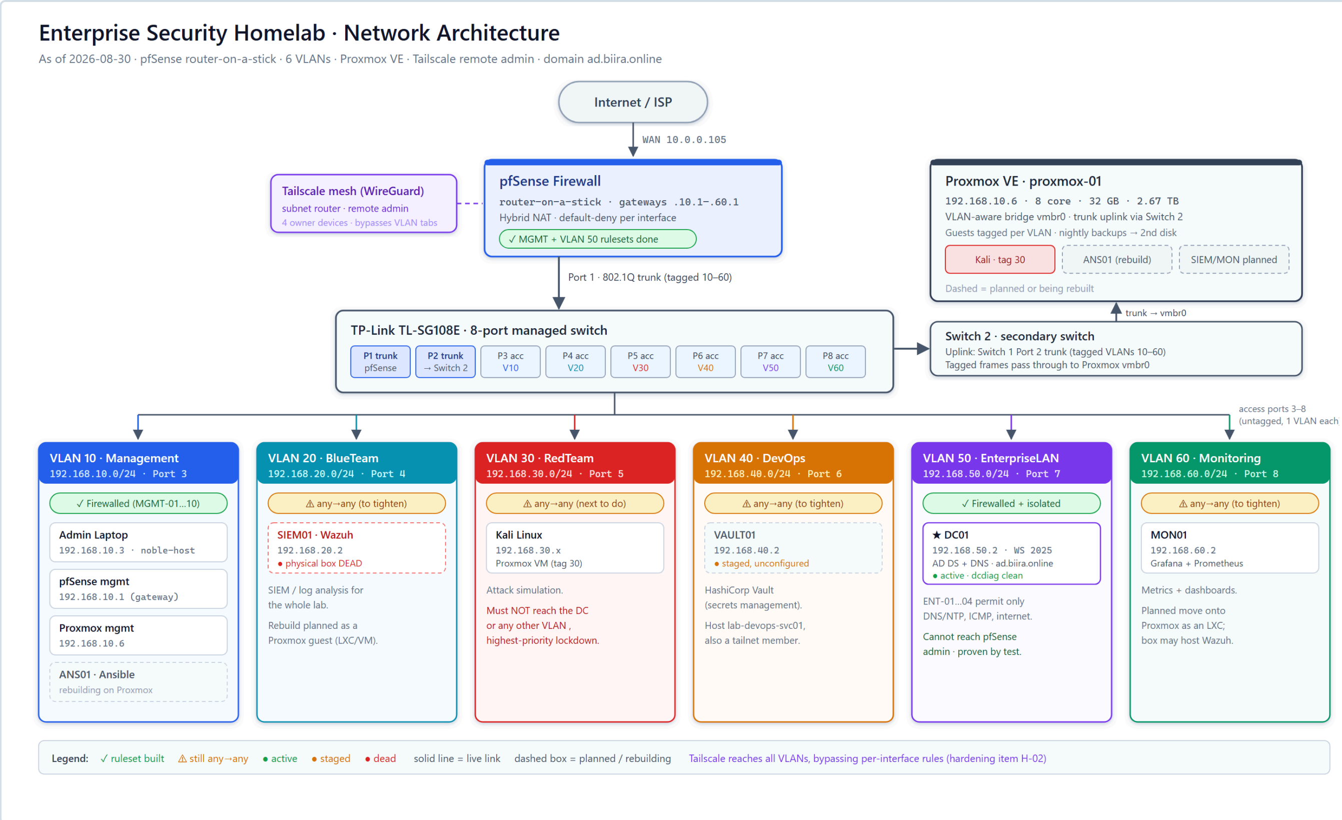
Internet ↔ Home Router ↔ pfSense (WAN/LAN) ↔ Managed Switch ↔ VLAN Segments
                                             ↕
                              All Lab Infrastructure
```

Network Architecture Diagram

Visual Overview



Current-state architecture (2026-08-30): Internet → pfSense → Switch 1 (trunk Port 1); Switch 1 Port 2 trunks to Switch 2, which carries the tagged VLANs to Proxmox; Ports 3–8 are single-VLAN access ports feeding the six security zones. Firewall status per zone is shown (green = ruleset built, amber = still any→any).

Interactive Architecture Diagram

For a detailed interactive view with hover effects and clickable elements, see the [Interactive Network Architecture Diagram]().

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**Interactive Features:**
- **Hover Effects**: Mouse over elements for enhanced visibility
- **Clickable VLANs**: Click on VLAN segments for detailed information
- **Live Configuration Data**: Real-time display of IP ranges and port assignments
- **Color-Coded Zones**: Visual distinction between security domains
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Architecture Highlights

- **pfSense Firewall**: Enterprise-grade routing and security with 2 NICs
- ** Managed Switch**: TP-Link TL-SG108E with 8 ports supporting VLAN tagging
- **VLAN Segmentation**: 6 isolated network zones for different security functions
- **Security Model**: Default-deny firewall with explicit rules for each VLAN
- **Hybrid NAT**: Controlled outbound access for all network segments

pfSense Configuration

Interface Configuration

Base Interface Changes

- **Default LAN (ue0)**: Changed from `192.168.1.1` to `192.168.99.1`
- **Isolation Applied**: No DHCP, blocked traffic for security
- **VLAN Parent**: All VLANs created under interface `ue0`
- **WAN Interface**: Maintains connection to upstream router

Interface Assignment Summary

Interface	Type	Purpose	IP Address
WAN	Physical	Internet connectivity	DHCP from ISP
LAN (ue0)	Physical	VLAN parent interface	`192.168.99.1` (isolated)
VLAN 10	Virtual	Management access	`192.168.10.1`
VLAN 20	Virtual	BlueTeam security	`192.168.20.1`
VLAN 30	Virtual	RedTeam operations	`192.168.30.1`
VLAN 40	Virtual	DevOps pipeline	`192.168.40.1`
VLAN 50	Virtual	EnterpriseLAN	`192.168.50.1`
VLAN 60	Virtual	Monitoring stack	`192.168.60.1`

All six VLAN interfaces are **802.1Q tagged sub-interfaces of the single physical LAN NIC `ue0`**. This is the pfSense half of a *router-on-a-stick* design: one cable to the switch trunk carries every VLAN, and pfSense does the routing between them.

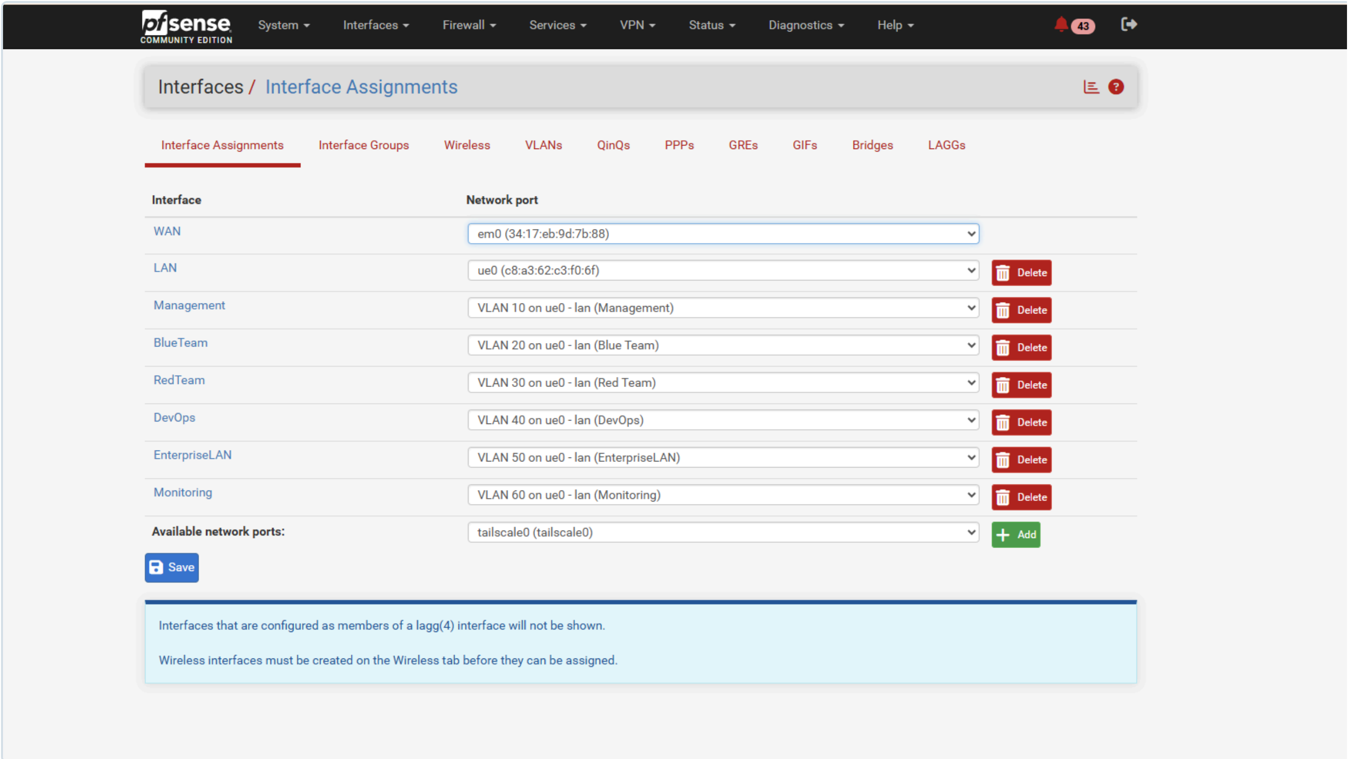


Figure 1.1: Interfaces → Assignments. Each VLAN (Management 10 … Monitoring 60) is a tagged sub-interface of the parent `ue0`, alongside WAN (`em0`) and the isolated LAN. This is the pfSense half of the 802.1Q trunk; the switch half appears in Figures 1.4–1.5.

Each VLAN interface has a static gateway IP and **no upstream gateway** (it is a local network, not an internet uplink), so pfSense treats it as a routed LAN segment:

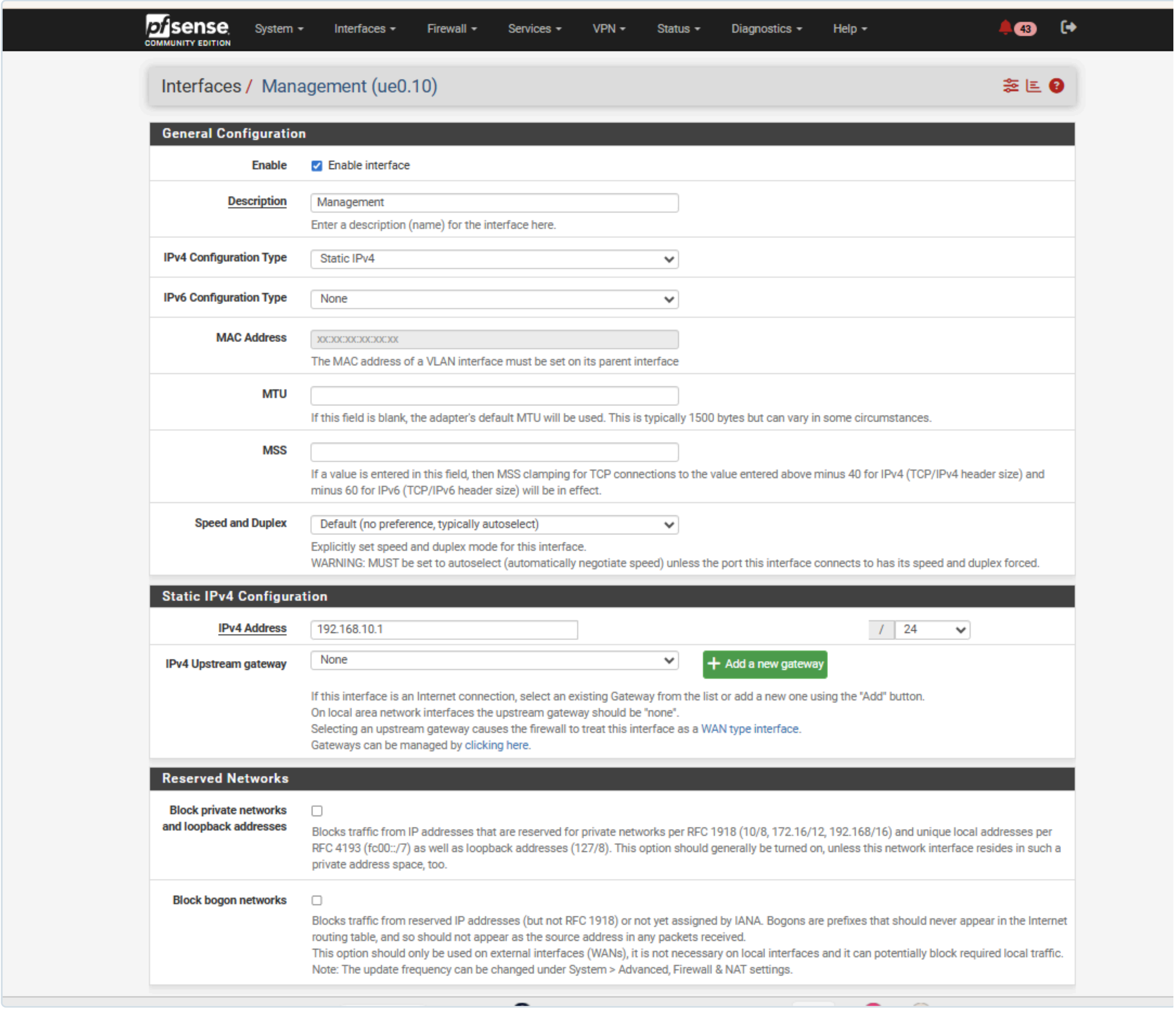


Figure 1.2: Example interface detail (Management, `ue0.10`): static IPv4 `192.168.10.1/24`, upstream gateway set to None because inter-VLAN routing is handled internally by pfSense. The same pattern is applied to VLANs 20–60.

VLAN & Subnet Architecture

Design Philosophy

The network implements **VLAN-based segmentation** where each VLAN represents a separate logical security zone. This approach ensures: - **Layer 2 isolation** between different environments - **Granular firewall control** via pfSense rules - **Enhanced monitoring** and policy enforcement - **Clear separation** between Blue Team and Red Team activities

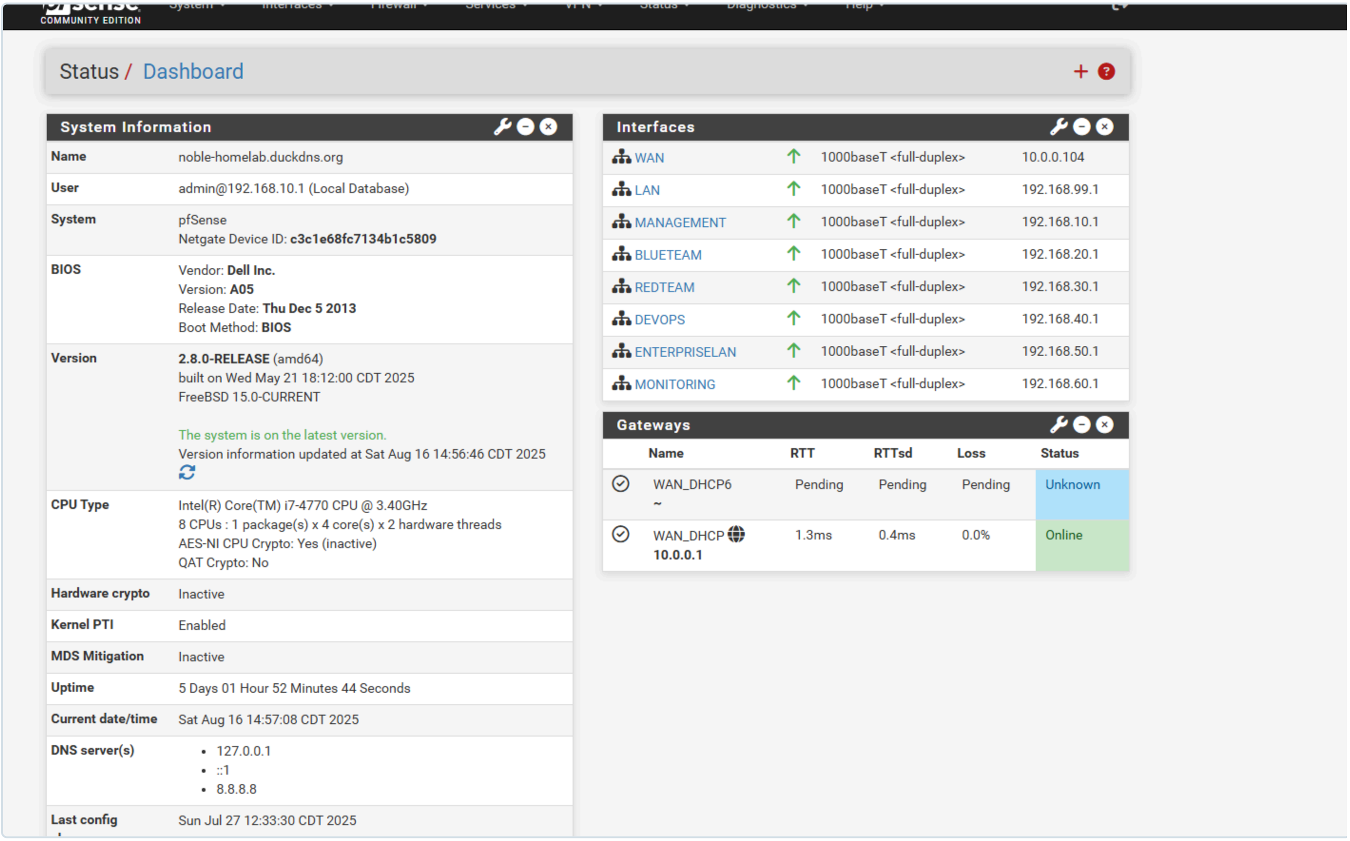


Figure 1.3: pfSense dashboard: all six VLAN interfaces up, each with its gateway IP (`192.168.10.1` … `192.168.60.1`), plus the WAN uplink. This is the live confirmation that every segment specified below is active and routing.

Complete VLAN Specification

Management	`10`	`192.168.10.0/24`	`192.168.10.1`	Administrative access to pfSense Web UI and management systems. Restricted to trusted admin devices only.
BlueTeam	`20`	`192.168.20.0/24`	`192.168.20.1`	Security monitoring infrastructure including Wazuh SIEM, ELK stack, IDS sensors, and defense tools.
RedTeam	`30`	`192.168.30.0/24`	`192.168.30.1`	Isolated environment for attack simulation, penetration testing, and offensive security experiments.
DevOps	`40`	`192.168.40.0/24`	`192.168.40.1`	CI/CD infrastructure, build servers, deployment automation, and DevSecOps toolchain.
EnterpriseLAN	`50`	`192.168.50.0/24`	`192.168.50.1`	Simulated business services including internal DNS, web applications, and corporate server infrastructure.

Monitoring	`60`	`192.168.60.0/24`	`192.168.60.1`	Dedicated observability stack hosting Prometheus, Grafana, Loki, and out-of-band monitoring systems.

Network Segmentation Benefits

- **Enhanced Security**: Layer 2 isolation with pfSense firewall control
- **Clear Visibility**: Each team/function has dedicated network space
- **Safe Testing**: Attack simulations contained within RedTeam VLAN
- **Monitoring**: Centralized observability without network interference

DHCP Configuration Strategy

IP Address Allocation Plan

Each VLAN subnet uses a structured IP allocation to prevent conflicts and ensure predictable addressing:

- **Static Range**: `.2` through `.49` (48 addresses for servers/infrastructure)
- **DHCP Pool**: `.50` through `.100` (51 addresses for dynamic assignment)
- **Reserved**: `.101` through `.254` (154 addresses for future expansion)

DHCP Scope Configuration

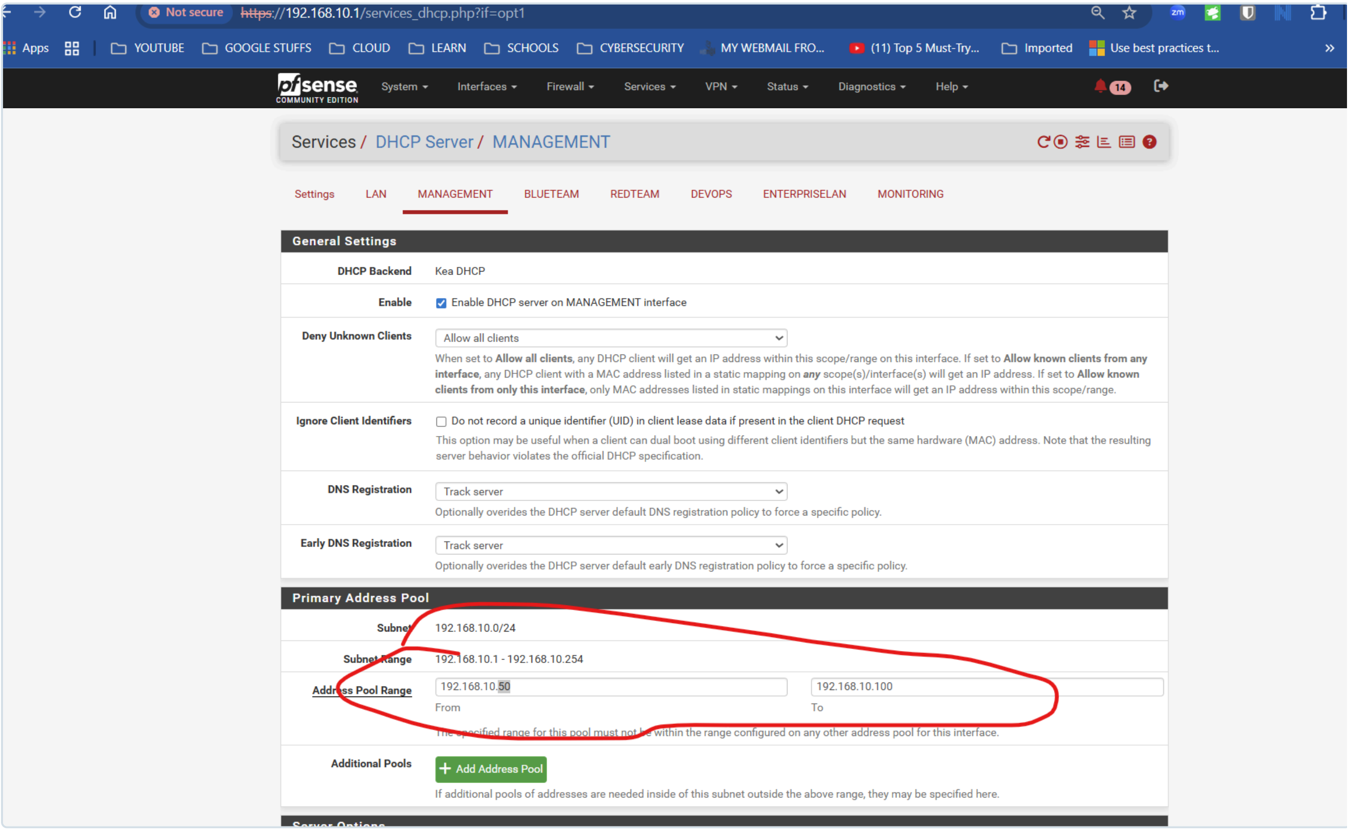
Example: Management VLAN (192.168.10.0/24)

- **Static Assignments**: `192.168.10.2` – `192.168.10.49`
- **DHCP Dynamic Pool**: `192.168.10.50` – `192.168.10.100`
- **Future Use**: `192.168.10.101` – `192.168.10.254`

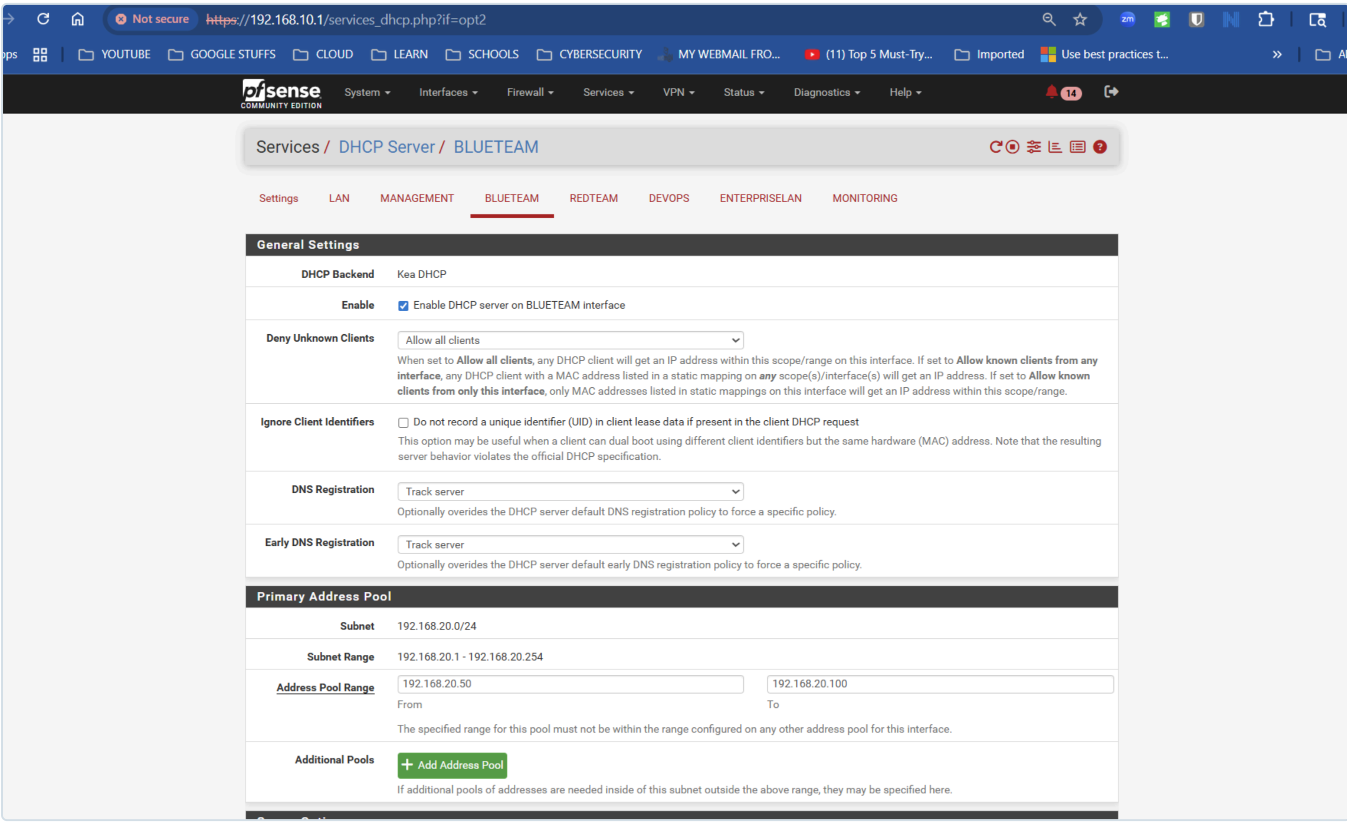
This pattern is replicated across all VLANs for consistency.

DHCP Implementation Evidence

Management VLAN DHCP Configuration



BlueTeam VLAN DHCP Configuration



Strategic Benefits

- ✅ **Predictable Infrastructure**: Critical systems use static IPs
- ✅ **Conflict Prevention**: Clear separation between static and dynamic ranges
- ✅ **Scalability**: Significant room for future growth
- ✅ **Consistency**: Identical pattern across all VLANs

Firewall Rules & Access Control

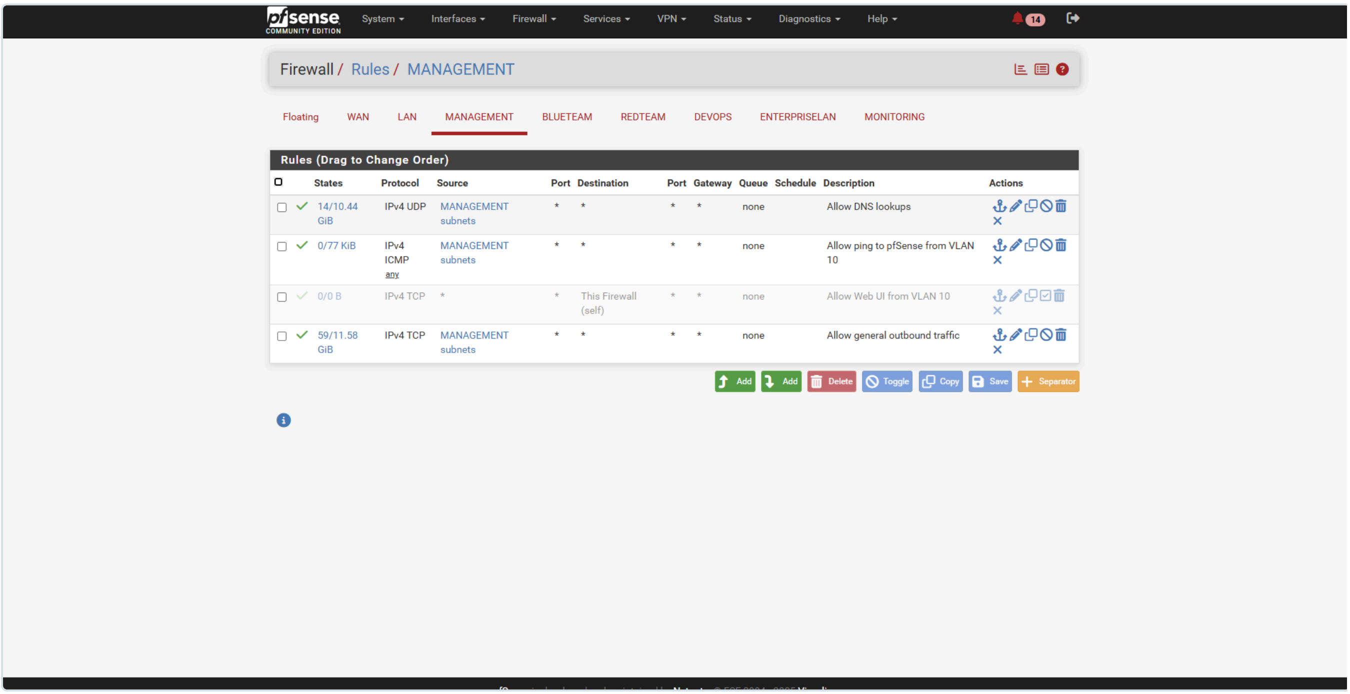
Security Model

pfSense implements a **default-deny** security model where all traffic is blocked unless explicitly allowed. Each VLAN has customized firewall rules based on its security requirements and operational needs.

Management VLAN (VLAN 10) Rules

Action	Protocol	Source	Destination	Purpose
Allow	ICMP	VLAN 10 Subnet	Any	Network troubleshooting and connectivity testing
Allow	TCP/UDP	VLAN 10 Subnet	This Firewall	Access to pfSense Web UI (HTTPS)
Allow	UDP	VLAN 10 Subnet	Any (Port 53)	DNS resolution services
Allow	Any	VLAN 10 Subnet	Any	Full internet access for administrative tasks

Management VLAN Firewall Implementation

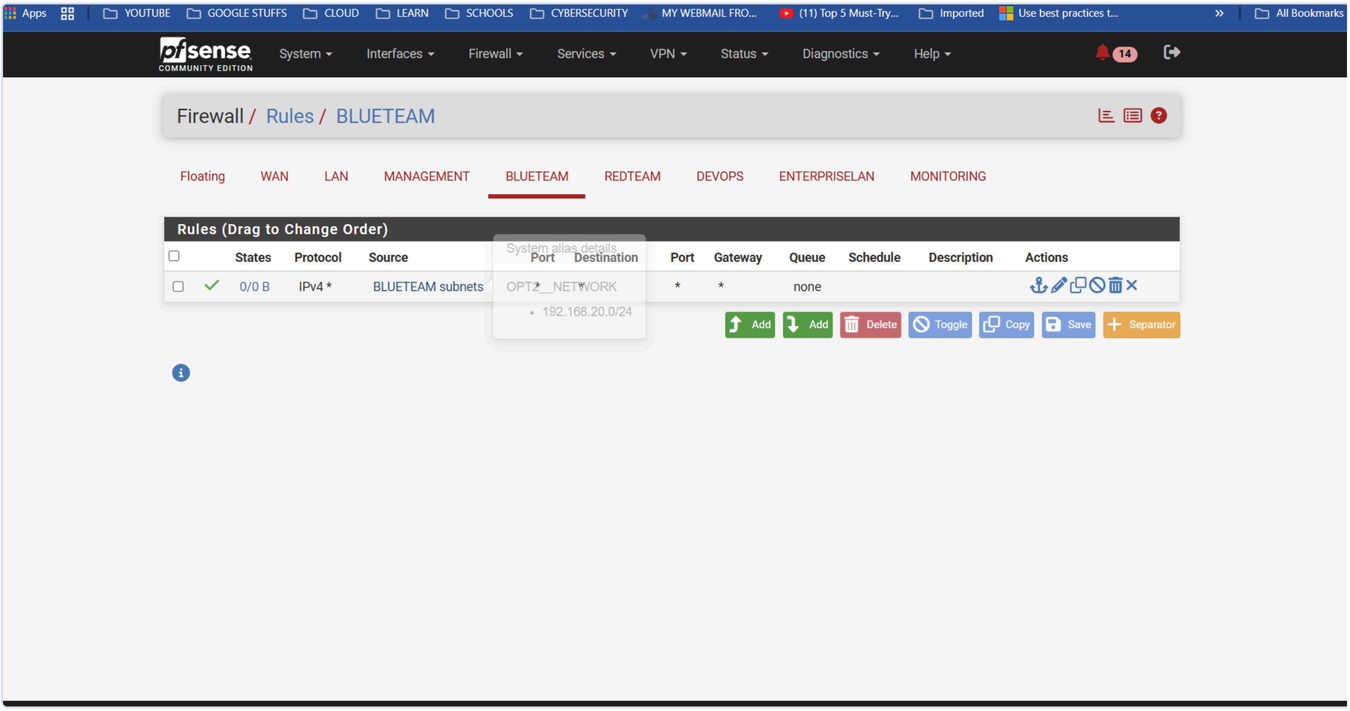


Other VLANs Security Rules

For **BlueTeam, RedTeam, DevOps, EnterpriseLAN, and Monitoring** VLANs:

Action	Protocol	Source	Destination	Purpose
Allow	Any	VLAN [X] Subnet	Any	Full internet access for operational requirements

Additional VLANs Firewall Configuration



Firewall Rule Strategy

- **Management VLAN**: Most permissive with administrative access
- **Operational VLANs**: Internet access with potential for future restrictions
- **Inter-VLAN Communication**: Controlled by specific rules as needed
- **Default Behavior**: All traffic denied unless explicitly permitted

Active Directory Firewall Aliases

Why aliases

A pfSense **alias** is a named list of hosts or ports that a firewall rule can reference by name. Instead of eight separate rules ("allow TCP 53 to the DC", "allow TCP 88 to the DC", ...) there is one rule that reads "allow `AD_TCP` to `SRV1_DC`". Rules become self-describing, and when a port needs adding or removing it is changed once in the alias rather than in every rule that uses it.

Active Directory is not a single service. A domain member relies on a bundle of protocols (DNS, Kerberos, LDAP, SMB, RPC, NTP) that all terminate on the domain controller. Blocking any one of them produces confusing half-working behaviour (can join but cannot log on, logs on but Group Policy fails, password changes error out). The aliases below group those protocols by role so that access can be granted precisely.

Alias inventory

Alias	Type	Contents	Purpose
`SRV1_DC`	Host	`192.168.50.2`	Windows Server 2025 domain controller `dc01.ad.biira.online` (alias name predates the host rename SRV1→DC01)
`AD_TCP`	Ports	53, 88, 135, 389, 445, 464, 636, 3268	TCP ports a domain member needs on the DC
`AD_UDP`	Ports	53, 88, 123, 389, 464	UDP ports a domain member needs on the DC
`AD_RPC_DYNAMIC`	Ports	`49152:65535`	RPC dynamic high-port range (see note)
`MGMT_TCP`	Ports	3389, 5985, 5986	Remote administration only: RDP and WinRM

`AD_TCP`: domain-membership TCP ports

Port	Protocol	What it does
53	DNS	Clients look up SRV records such as `_ldap._tcp.ad.biira.online` to locate the DC. TCP carries large answers and zone transfers; UDP carries ordinary queries.
88	Kerberos	The core AD authentication protocol. Every logon and every access to a network resource obtains a Kerberos ticket from the DC.
135	RPC endpoint mapper	The "directory assistance" for Microsoft RPC. A client asks 135 which dynamic port a given service is listening on, then connects there. Required for domain join, Group Policy and remote management.
389	LDAP	Reading and writing the directory: user lookups, group membership, computer objects.
445	SMB	File sharing. Clients pull Group Policy objects and logon scripts from the DC's `SYSVOL` and `NETLOGON` shares.
464	Kerberos password change	Used when a user changes a password or a computer rotates its machine-account password (every 30 days by default).
636	LDAPS	LDAP over TLS. Not required by Windows itself, but Linux tooling, HashiCorp Vault's LDAP auth method and Wazuh integrations prefer it. Included so those work without a later rule change.
3268	Global Catalog	A read-only, forest-wide index of the directory. Logon uses it to resolve universal group membership; directory-aware applications search it.

`AD_UDP` : domain-membership UDP ports

Port	Protocol	What it does
53	DNS	Everyday name resolution.
88	Kerberos	Kerberos tries UDP first for small requests and falls back to TCP.
123	NTP	Domain members sync their clocks to the DC. Kerberos rejects tickets when clocks differ by more than 5 minutes, so this port is silently critical.
389	LDAP ping	A lightweight "are you a DC for my site?" probe that clients send while locating a domain controller.
464	Kerberos password change	UDP variant of the password-change service.

`AD_RPC_DYNAMIC` : why it is separate

After a client asks port 135 where a service lives, the DC answers with a random port in the **49152 to 65535** range and the client connects there. Domain join, Group Policy processing and AD replication all depend on this. Because the range is large it is kept in its own alias and granted only from tightly scoped sources (Management VLAN to `SRV1_DC`), never bundled into `AD_TCP` where it might be handed out more broadly.

`MGMT_TCP` : why RDP and WinRM are not in `AD_TCP`

RDP (3389) and WinRM (5985 plain, 5986 TLS) are for *administering* the server, not for *being a member* of its domain. Keeping them separate means that granting a client VLAN " `AD_TCP` to `SRV1_DC` " for domain membership never accidentally grants remote console or automation access as well. `MGMT_TCP` is granted only from the Management VLAN (administrator laptop and the Ansible controller).

Outbound NAT Configuration

NAT Implementation Approach

To enable proper internet connectivity for all VLAN segments, pfSense uses **Hybrid Outbound NAT** mode, providing explicit control over network address translation while maintaining automatic rule generation for standard interfaces.

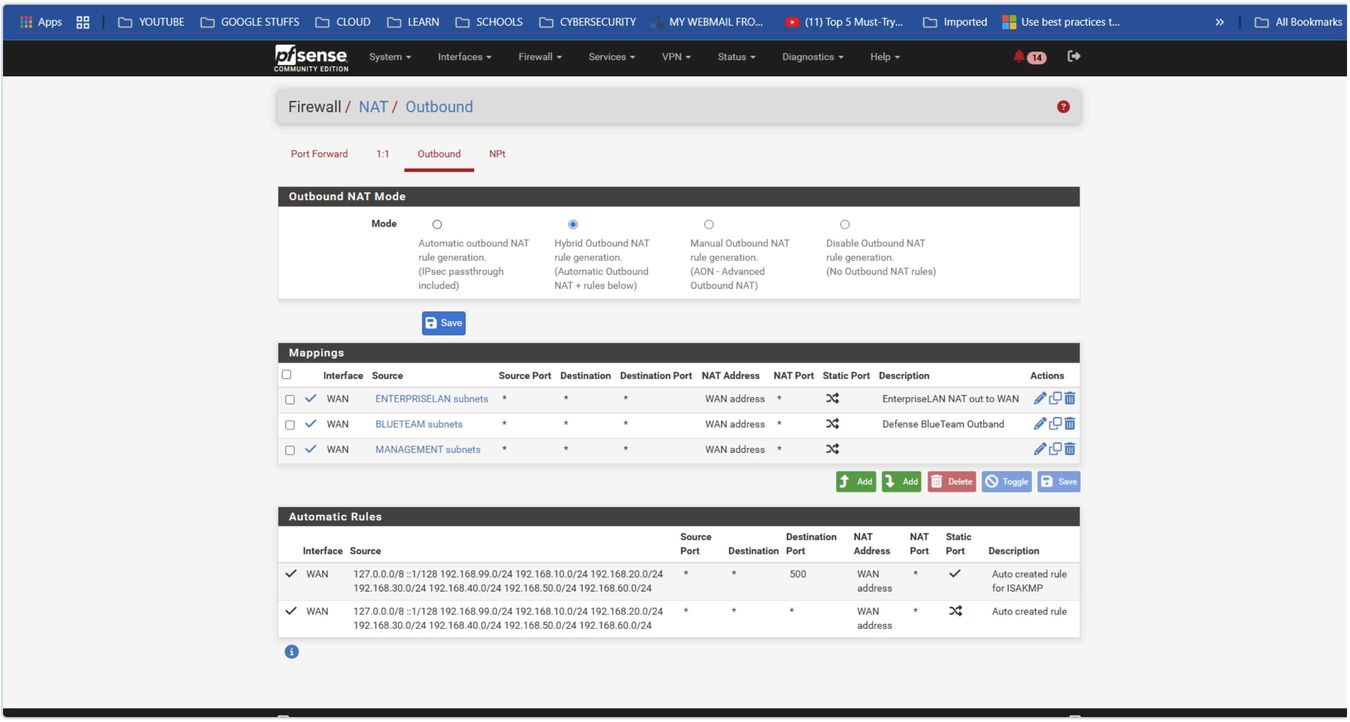
Configuration Method

1. **Mode Selection**: **Firewall > NAT > Outbound**
2. **Mode Change**: Switched to **"Hybrid Outbound NAT rule generation"**
3. **Rule Creation**: Manual rules defined for each VLAN subnet

NAT Rule Specification

VLAN Name	Source Subnet	Translation Target	Status
Management	`192.168.10.0/24`	WAN Interface	Active
BlueTeam	`192.168.20.0/24`	WAN Interface	Active
RedTeam	`192.168.30.0/24`	WAN Interface	Active
DevOps	`192.168.40.0/24`	WAN Interface	Active
EnterpriseLAN	`192.168.50.0/24`	WAN Interface	Active
Monitoring	`192.168.60.0/24`	WAN Interface	Active

Hybrid Outbound NAT Configuration



Hybrid NAT Advantages

Flexibility: Manual control over VLAN-specific NAT behavior
Compatibility: Retains pfSense default rules for other interfaces
Scalability: Easy to modify or restrict specific VLANs
Visibility: Clear understanding of NAT translations

Switch Port Configuration

TP-Link TL-SG108E Port Mapping

The managed switch provides VLAN segmentation through strategic port assignments, supporting both trunk and access port configurations.

Port 1	Trunk (Tagged)	**All VLANs**: 10,20,30,40,50,60	pfSense `ue0` interface connection
Port 2	Trunk (Tagged)	**All VLANs**: 10,20,30,40,50,60	Trunk to **Switch 2** (secondary switch). Proxmox VE (`192.168.10.6`) connects via Switch 2 and receives the tagged VLANs for multi-VLAN VM hosting
Port 3	Access (Untagged)	**VLAN 10 Only**	Management VLAN direct access
Port 4	Access (Untagged)	**VLAN 20 Only**	BlueTeam VLAN direct access
Port 5	Access (Untagged)	**VLAN 30 Only**	RedTeam VLAN direct access
Port 6	Access (Untagged)	**VLAN 40 Only**	DevOps VLAN, dedicated HashiCorp Vault desktop (`192.168.40.2` , awaiting configuration)
Port 7	Access (Untagged)	**VLAN 50 Only**	EnterpriseLAN VLAN, Windows Server 2025 domain controller (`192.168.50.2` , via USB-to-Ethernet adapter)
Port 8	Access (Untagged)	**VLAN 60 Only**	Monitoring VLAN direct access

Switch Port Configuration Evidence

The switch enforces the same VLAN scheme from the Layer 2 side. Two settings define it: **802.1Q VLAN membership** (which ports carry which VLANs, and whether tagged or untagged) and the **PVID** (the VLAN an *untagged* frame is placed into as it enters a port).

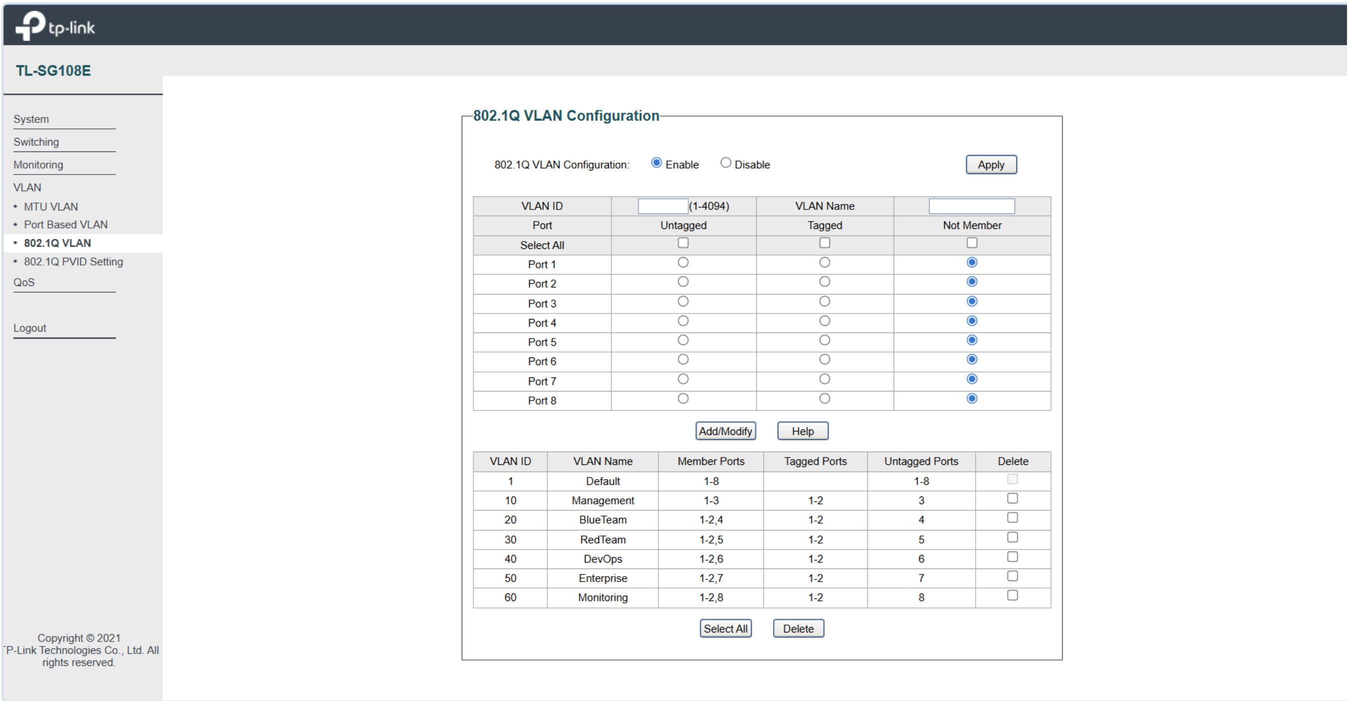


Figure 1.4: TP-Link 802.1Q VLAN table. Every VLAN (10–60) includes ports 1–2 as tagged (the trunk uplinks to pfSense and Proxmox) plus one untagged access port (10→3, 20→4, 30→5, 40→6, 50→7, 60→8). Tagged keeps the 802.1Q VLAN ID on the frame so a trunk can carry many VLANs; untagged strips it for a plain end device that knows nothing about VLANs.

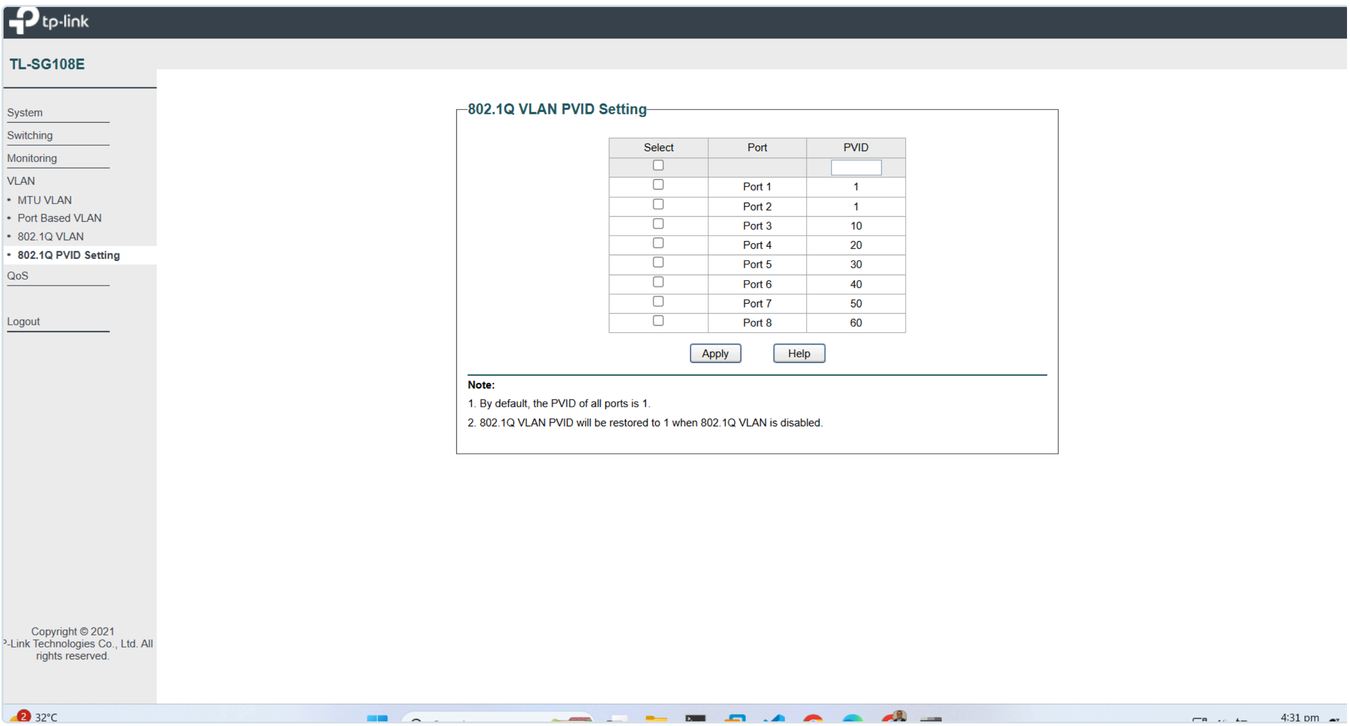


Figure 1.5: 802.1Q PVID settings. Each access port 3–8 carries the PVID of its VLAN (10, 20, 30, 40, 50, 60), so any untagged device plugged in lands in the correct segment automatically. Trunk ports 1–2 keep PVID 1 because they only ever carry tagged traffic. Figures 1.4 and 1.5 together are the switch-side proof of the port map in the table above, and they mirror the pfSense sub-interfaces in Figure 1.1.

Port Configuration Strategy

- **Trunk Ports (1-2)**: Carry all tagged VLAN traffic, Port 1 uplinks pfSense; Port 2 trunks to Switch 2, which carries the tagged VLANs on to the Proxmox VE VLAN-aware bridge
- **Access Ports (3-8)**: Provide direct, untagged access to specific VLANs
- **Device Placement**: End devices automatically assigned to appropriate VLAN
- **Scalability**: Additional devices easily added to any VLAN segment

Security Implementation

Network Security Measures

- **LAN Interface Isolation**: Original `ue0` interface secured with no DHCP or routing
- **Administrative Access Control**: All pfSense GUI access restricted to Management VLAN
- **Inter-VLAN Security**: No default communication between VLANs
- **Controlled Connectivity**: ICMP enabled only for troubleshooting purposes

Access Control Summary

- **pfSense Management**: Only accessible via `http://192.168.10.1` on VLAN 10
- **VLAN Isolation**: Each VLAN operates independently unless rules permit interaction
- **Firewall Protection**: All traffic subject to pfSense security rules
- **Default Deny**: Implicit denial of all traffic not explicitly permitted

Implementation Challenges & Solutions

Key Issues Resolved

Issue 1: Lost Administrative Access

Problem: Disabling original LAN interface caused loss of pfSense access
Solution: Assigned fallback IP and implemented blocking instead of disabling
Lesson: Always maintain administrative access during network changes

Issue 2: Windows Client Connectivity

Problem: Windows clients unable to ping across VLANs
Solution: Enabled ICMP Echo in pfSense firewall rules
Resolution: Network troubleshooting capabilities restored

Issue 3: Internal vs External Connectivity

Problem: External connectivity worked (8.8.8.8) but internal failed
Root Cause: Local firewall rules blocking internal VLAN communication
Fix: Adjusted firewall rules for appropriate inter-VLAN access

Issue 4: GUI Access Verification

Problem: Uncertainty about pfSense management access
Verification: Confirmed GUI accessible via `http://192.168.10.1` on VLAN 10
Result: Administrative access properly secured and functional

Implementation Verification

Network Connectivity Tests

- **VLAN Gateway Access**: All VLANs can reach their respective gateways
- **Internet Connectivity**: Outbound NAT functional for all segments
- **DNS Resolution**: Name resolution working across all VLANs
- **Administrative Access**: pfSense GUI accessible from Management VLAN

Security Verification

- **VLAN Isolation**: Confirmed separation between network segments
- **Firewall Rules**: All configured rules operational and effective
- **Access Controls**: Administrative functions properly restricted
- **NAT Translation**: Proper address translation for internet access

Next Phase Integration

Prepared Infrastructure

The network foundation now supports: - **Security Services**: SIEM deployment in BlueTeam VLAN - **Monitoring Systems**: Observability stack in Monitoring VLAN - **Automation Platforms**: Ansible controller in Management VLAN - **Testing Environments**: Red Team tools in isolated RedTeam VLAN

Expansion Capabilities

- Additional VLANs easily configured
- Scalable DHCP and firewall rule management
- Support for complex inter-VLAN communication requirements
- Ready for enterprise service deployment

Network Infrastructure Status: Complete and Operational
Next Phase: [Security Monitoring Stack Deployment]